



Bayesian Models for Spatial Equity: Separating Latent Demand from Supply in EV Charging Infrastructure

Demand that has never been built for is never observed. This study proposes a hierarchical framework that separates **latent charging demand** from the **supply filter** through which it is recorded — and propagates that uncertainty into the siting decision itself. Electric vehicle charging infrastructure in West Java, Indonesia.

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THE ARGUMENT IN ONE LINE — FROM UNDER-DETECTED RECORD TO SITING DECISION

WHAT IS RECORDED

Sessions y — an under-detected count

Only demand that met a charger becomes data — what the record shows is demand already filtered by supply.

WHAT IS MODELLED

Capture κ demand d

The supply filter and the demand behind it, separated by supply-side covariates that move only κ .

WHAT IS RECOVERED

A posterior for every area

Latent demand with honest uncertainty — widest where the record is thinnest.

WHAT IT DECIDES

$P(\text{unmet demand} > 1)$

Sites ranked by the probability a build is viable — and equity indices that keep their width.

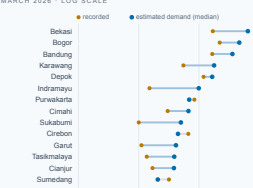
→ 01

The observation is conditional on the supply

Charging infrastructure is sited where utilisation is already high. But utilisation is only recorded where a charger exists. The planner's evidence base is a sample selected by the very decision it is meant to inform.

The result is self-reinforcing: capacity flows to districts that already have it, latent demand elsewhere stays invisible because it never becomes a session, and equity is then assessed on the same under-detected record — which makes demand look far more concentrated than it is, because low-provision areas cannot generate the sessions that would reveal them.

RECORDED SESSIONS VS ESTIMATED DEMAND, BY AREA - MARCH 2026 - LOG SCALE



Estimated demand exceeds the recorded count in eleven of fourteen areas, by factors up to 6.6x (Indramayu). Purwakarta, Cirebon and Sumedang sit the other way round: recorded activity exceeds local estimated demand, because through-traffic charges on toll corridors registered to those districts. Neither pattern is visible in the operator record alone.

→ 02

Why this problem is Bayesian, not merely fitted

- Sparsity has no maximum-likelihood answer.** Most units record zero sessions because they hold zero chargers; area-level rates are undefined or widely unstable. Hierarchical priors borrow strength across neighbours instead of discarding the unit.
- Spatial structure is a parameter, not a nuisance.** The BYM2 reparameterisation splits structure from unstructured heterogeneity with a single intermixing parameter ρ , under a prior we can actually defend.
- The decision needs a probability, not an estimate.** A siting committee should ask $P(\text{unmet demand} > \text{viability threshold})$, which the posterior returns directly. A point estimate with a standard error does not answer it.
- Inequality indices are non-linear functionals.** Gini and Lorenz computed on estimated rates have no clean analytic standard error. Posterior draws propagate uncertainty through them exactly — see panel 06.

→ 03

Model: demand and capture, estimated jointly

The canonical imperfect-detection (**N-mixture**) structure: observed sessions y_i are latent demand d_i thinned by a capture probability $a_i \in (0,1)$, plus a corridor inflow c_i for non-resident through-traffic — so the empirical ratio recorded $\div$ latent can exceed 1 where c_i is large, while a_i itself never does.

OBSERVATION

$$y_i \sim \text{NegBin}(\mu_i, \phi), \quad \mu_i = d_i \cdot a_i + c_i$$

c_i = corridor inflow: non-resident through-traffic, nonzero only in toll-corridor districts

LATENT DEMAND

$$\log d_i = \beta_0 + x_i^T \beta + u_i + v_i$$

registered EV fleet (primary), home-charger uptake (substitution, negative), population & activity density, trip generation, income, POI mix

CAPTURE / AVAILABILITY FILTER

$$\log a_i = \alpha_0 + x_i^T \alpha + z_i^T \gamma$$

supply-side only: distance to MV substations with spare capacity, parcel availability, resident drive-time to the nearest charger — network topology, not driver preference

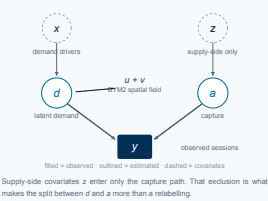
SPATIAL PRIOR — BYM2

$$u + v = \tau^{-1/2} (\sqrt{\rho} \cdot u' + \sqrt{1-\rho} \cdot v')$$

$$u' \sim \text{ICAR}(\mathbf{W}), v' \sim \text{N}(\mu, \Sigma), \text{PC priors on } \tau \text{ and } \rho: \rho \sim \text{Beta}(\frac{a}{2}, \frac{b}{2})$$

Model variants are iterated with **INLA**; every figure reported on this poster comes from the **MCMC fit** (Stan NUTS, 24,000 iterations, 1,000 retained draws). The BYM2 spatial term is specified for the sub-district stage.

DIRECTED STRUCTURE



RESEARCH DESIGN — FIVE STAGES

- Specification** — joint demand-capture model with BYM2 spatial prior at sub-district level.
- Calibration before data** — prior predictive simulation and simulation-based calibration on synthetic data with known d .
- Estimation** — fitted with INLA across model variants; preferred specification validated against Stan (NUTS).
- Checking** — posterior predictive checks on the zero mass; PSIS-LOO against baseline models; sensitivity analysis over ρ priors.
- Decision outputs** — posteriors for unmet demand, exceedance-probability site ranking, and equity indices with credible intervals.

→ 04

The identification problem, stated plainly



Nothing in the data distinguishes low demand, well served from high demand, barely served: both produce few sessions. Separating d_i from a_i requires an argument, not more data.

The argument is an exclusion restriction. Grid headroom at the nearest medium-voltage substation and land parcel availability determine where a charger can physically be built; they carry no information about whether drivers in that district want to charge. They move a and not d .

This assumption is the weakest link and will be treated as such: prior sensitivity analysis over γ , simulation-based calibration on synthetic data with known d , and a falsification check on districts where supply expanded for reasons unrelated to demand.

HOW THE MODEL EARNS TRUST

Recovers known d from synthetic data

Beats a naive Poisson fit on held-out districts

Posterior predictive replicates the zero mass

Conclusions survive alternative γ priors

SBC RANKS

PPC

SENSITIVITY

THREATS TO IDENTIFICATION, AND HOW EACH IS MET

- Reverse causality** — Supply may follow demand. The instrument uses grid headroom and parcel availability — fixed by legacy network topology, not by drivers — so it shifts capture without tracking demand.
- Through-traffic** — Toll corridor stations serve non-residents: Purwakarta, Cirebon and Sumedang record more sessions than local demand (gap ratio > 1). A corridor flag enters the capture equation, not the demand equation.
- Prior sensitivity** — Capture is weakly identified where supply is thin; the fit is repeated under alternative γ priors and conclusions are reported only where they survive.
- Aggregation bias** — Service-area totals mask internal gaps; finer disaggregation with a BYM2 spatial prior is the next stage, with these results as its benchmark.

LIMITATIONS, AND HOW THEY ARE ADDRESSED

- Results can be prior-sensitive — sensitivity analysis over alternative priors is built into the design.
- Identification rests on the exclusion restriction — treated as the weakest link and falsification-checked.
- Computationally heavier than a single fit — INLA for iteration speed, Stan reserved for validation.
- Conclusions are conditional on model specification — predictive checks replicate the data's key features.
- Single-month fit, 14 units, <13 parameters — monthly replication Jan 2024-Jun 2026 (432 area-months) is the identifying next stage; temporal replication is what pins down N-mixture models.



Live data dashboard
The March 2026 records, equity indices and site-selection model behind this poster — [jabar-ev-nutcracker](#) app.

→ 05

Setting and data — West Java

West Java — 27 cities and regencies, ~50 million residents, the most active front of Indonesia's public-charging rollout. The March 2026 record resolves to 14 operator service areas (41.1 million residents, 633 charger units), the level at which sessions are reported. Analysis is at service-area level; finer disaggregation with a 1 km grid is the next stage.

LAYER	SOURCE
Charger locations, power rating, operator, status	Master station registry, West Java utility, Mar 2026
Transaction detail: kWh, duration, revenue — sets κ	101,020 sessions, Mar 2026
EV-owner home-charging installations by sub-district	customer applications, Nov 2025 — Jun 2026
Service-quality perception by station — next stage	review-based sentiment, 584 stations
Road network, points of interest, trip proxies	OpenStreetMap
Population (census 2022), HDI 2023, income tier	national statistics agency

All quantiles on this poster are computed from the actual West Java network records: 101,020 transactions and 2.26 GWh (March 2026 extract), 633 charger units, and census population. Customer-level records were aggregated to area level and anonymized, used under data agreement with the utility. Posterior figures come from the MCMC fit (Stan NUTS, 24,000 iterations, 1,000 retained draws). Sub-district disaggregation and the BYM2 spatial term are the next stage.

Three quantiles, one dataset

MEASURED	ESTIMATED	DERIVED
Observed sessions recorded, but under-detected: thinned by supply	Latent demand the target — never directly observed	$d_i - \kappa$ unmet demand what siting decisions should be ranked on

101,020 sessions recorded March 2026
229,000 latent demand posterior median
***2.3** gross-area-wide demand gap

The three quantiles are not interchangeable, and the distinction is the whole argument. A ranking built on $d_i - \kappa$ answers "where has charging already happened?", a ranking built on d_i — κ answers "where would charging happen if capacity existed?". Only the second is a forward-looking investment criterion, and only a model that estimates the capture filter can compute it.

WHO ACTS ON WHICH QUANTITY

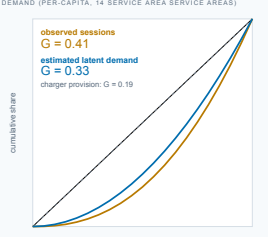
- Operator KPI** — y_i — for throughput and revenue of the existing fleet; correct for operations, independent of expansion.
- Equity audit** — d_i — need as it exists, served or not; the only basis on which distributive fairness can be judged.
- Siting committee** — $d_i - \kappa$ — the probability a new unit clears viability, with estimation uncertainty priced in.
- Regulator** — posterior Gini of d and of unmet demand — inequality claims that carry their credible intervals.

→ 06

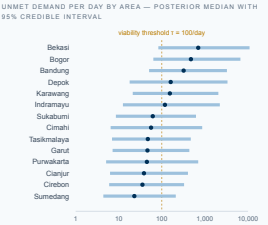
Equity indices that keep their uncertainty

Two distinct injustices must be kept apart: **access inequality** (supply concentrated in a few areas) and **unmet-demand inequality** (need left unserved) — only the second identifies where intervention changes welfare. Both are normally reported as a single Gini number; when the underlying quantity is estimated, the index is a posterior distribution, and its width is the difference between a defensible claim and a decorative one.

LORENZ CURVES — OBSERVED SESSIONS VS ESTIMATED LATENT DEMAND (PER-CAPITA, 14 SERVICE AREA SERVICE AREAS)



UNMET DEMAND PER DAY BY AREA — POSTERIOR MEDIAN WITH 95% CREDIBLE INTERVAL



Unmet charging events per day, log scale. Intervals are wide by construction: with an under-detected record, honest inference cannot be precise, and pretending otherwise is the error this design removes. The ordering of medians — not any single point — is the decision-relevant output, and it differs materially from the session ranking.

STRENGTHS OF THE BAYESIAN TREATMENT

- Structural zeros become information, not discarded noise — priors borrow strength across neighbours.
- Uncertainty propagates end-to-end, from data to Gini indices to site rankings.
- Delivers the decision quantity directly: $P(\text{unmet demand} > \text{viability threshold})$.
- Assumptions are explicit priors — auditable and testable, not implicit defaults.

→ 07

Conclusion: what the Bayesian treatment buys

Findings. Fitted on the March 2026 record — 101,020 sessions, 633 charger units, 14 service areas, 41.1 million residents — the model returns aggregate latent demand of **229,000 events against 101,020 recorded (*2.3 gap)**. Resident-demand capture spans 0.15–0.72 by area (aggregate 0.42 net of ~4,800 corridor-inflow sessions); gap ratios run 0.7x–6.6x. Observed sessions are more unequally distributed ($G = 0.41$) than estimated demand ($G = 0.33$). Capacity-adjusted unmet demand ($d - \kappa$, medians) ***2.260 events/day**, the raw demand-record gap is ***4,130/day**. And since installed capacity exceeds recorded use in most areas while capture stays low, the **binding constraint is accessibility — distance and siting — not capacity**.

Spatial equity. The three Gini indices order clearly: charger provision is the most even ($G = 0.19$), estimated demand less so ($G = 0.33$), observed sessions the most concentrated ($G = 0.41$). The concentration in the record is therefore generated by capture, not by where demand lives — toll corridors are overserved while populous inland areas go underserved. Reported at service-area resolution, these indices are a floor on the true inequity, never a ceiling.

Methodological. A joint demand-capture model with BYM2 spatial structure that turns an under-detected operator record into estimates of resident demand the record alone cannot show — with equity indices reported as posteriors rather than points.

Practical. Ranking candidate areas on raw sessions reproduces the existing map of supply; ranking them on $P(\text{unmet demand} > 1)$ does not. The two orderings differ materially: Indramayu rises from the bottom third of the session table to rank 6 and the investigate band on just 1,508 recorded sessions, while Purwakarta and Cirebon — busy in the record — drop out. The **pipeline runs end-to-end on real records: data in, posterior out, uncertainty intact, decision quantity delivered**.

FIT SUMMARY - MARCH 2026

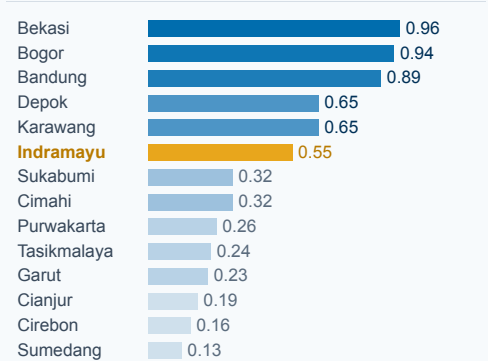
Areas - population - charger units	14 service areas (all 27 kabkota in scope) - 41.1M - 633
Sessions recorded — latent demand (median)	101,020 — 229,000
Capture κ (resident demand); aggregate - by area	0.42 - 0.15 - 0.72
Corridor inflow c (Purwakarta, Cirebon, Sumedang)	~4,800 sessions [†]
Gap ratio (latent ÷ recorded), range	0.67 - 6.59
Model - spatial prior - inference	NegBin demand-capture - MCMC (Stan NUTS)
Gini (per-capita, 14 service areas): provision - demand - sessions	0.19 - 0.33 - 0.41
Unmet demand $d - \kappa$ (capacity-adjusted) - raw gap - 31 days	*2,260 - *4,130 /day

[†] Corridor inflow: recorded use in these toll-corridor districts exceeds resident demand; it enters the observation model as a separate term c , keeping a a probability. Capacity $\kappa = 8 \times \text{event/day/tollroad}$ (mean session duration $\times$ operating window), so κ is a 5,120 deliverable event/day across 633 units.

RECOMMENDATIONS

- Site on unmet demand, not sessions.** Rank candidate areas by $P(\text{unmet demand} > 1)$; Indramayu rises to rank 6 and the investigate band on just 1,508 recorded sessions; corridor districts already at capture drop out.
- Audit equity on latent demand.** Regulators reporting session-based Gini overstate demand concentration and miss the structural gap; report the demand-based posterior ($G = 0.33$, per-capita alongside provision ($G = 0.19$) and sessions ($G = 0.41$), always labelling the service-area basis — aggregation makes these a floor on true inequity.
- Disaggregate before declaring coverage.** Georeference stations and sessions to kabupatenkota so gaps inside a service area become visible, and let each new build double as an experiment that narrows the posterior.
- Publish uncertainty with every index.** Fund decisions on credible intervals, and re-fit as each month's records arrive — the posterior is a living planning input, not a one-off study.

DECISION OUTPUT - $P(\text{UNMET DEMAND} > 100/\text{DAY})$



Posterior exceedance probabilities — the quantity an investment committee can act on directly. **Indramayu ranks sixth on unmet demand while recording only 1,508 sessions** — nowhere near the top of a session-based ranking. Purwakarta and Cirebon, busy in the operator record, fall to the lower half once capture is accounted for. Probabilities carry a Monte Carlo standard error of ± 0.02 — differences of one rank are not meaningful.

READING THE RANKING

- $P > 0.85$** Act now — unmet demand exceeds the viability threshold under almost every posterior draw.
- $0.5 - 0.85$** Investigate — viable in most draws; site-level survey resolves the remaining uncertainty.
- $0.3 - 0.5$** Monitor — the posterior is genuinely undecided; new monthly records move these areas fastest.
- $P < 0.3$** Defer — either genuinely low demand or already adequately served.

ABBREVIATIONS AND SYMBOLS

- Cri** — credible interval: the range holding 95% of posterior probability for a quantity
- [thr]** — viability threshold: minimum unmet demand (100 events/day) for a new site to be worth building
- ICAR** — intrinsic conditional autoregressive prior on the neighbourhood graph $\mathbf{W}$
- SBC** — simulation-based calibration: posterior predictive check on the zero mass; PSIS-LOO against baseline models; sensitivity analysis over ρ priors
- POI** — MV - OSM — points of interest - medium-voltage (grid) - OpenStreetMap

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